

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

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Code of Conduct:

I Athavan K(192524036)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Athavan K

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts

Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

1(a) Inductive Learning Approach, Target Variable & Key Features

Inductive learning is the process of deriving a general hypothesis (a predictive function $f: X \rightarrow Y$) from a finite set of labelled training examples, such that the hypothesis generalises well to unseen patients. For this healthcare dataset, supervised inductive learning is appropriate because every historical patient record already carries a confirmed diagnostic label (diabetic / non-diabetic), so the model can inductively learn the mapping from clinical features to outcome rather than discovering structure unsupervised.

Target variable: Diabetes_Status — a binary class label (1 = Diabetic, 0 = Non-Diabetic), confirmed via fasting glucose / HbA1c thresholds at the time the record was collected.

Key predictive features (with clinical and statistical justification):

- Glucose Level — directly measures blood-sugar regulation and is the clinical diagnostic criterion for diabetes (fasting glucose ≥ 126 mg/dL); expected to carry the highest mutual information with the label.
- BMI (Body Mass Index) — obesity is a well-established risk multiplier for Type-2 diabetes through insulin resistance; provides strong secondary discriminative power.
- Age — insulin sensitivity declines and cumulative metabolic risk rises with age, so age captures long-term risk accumulation not visible in a single glucose reading.
- Family History — encodes genetic predisposition; useful for early-onset cases where current biomarkers are still borderline.
- Blood Pressure / Cholesterol — co-morbid indicators of metabolic syndrome, useful as supporting (not primary) evidence.

1(b) Handling Class Imbalance

Diabetes datasets are typically imbalanced (commonly 70:30 or 80:20 non-diabetic : diabetic), which biases a naively-trained classifier toward the majority class and inflates accuracy while silently failing on the medically critical minority class.

Recommended strategy: SMOTE (Synthetic Minority Over-sampling Technique), applied only on the training fold, combined with class-weighting

- SMOTE generates synthetic diabetic-class samples by interpolating between a minority sample and its k-nearest minority neighbours in feature space, rather than

duplicating rows (as naive over-sampling does), so the model sees more diverse minority-class patterns and is less prone to overfitting on repeated points.

- Under-sampling the majority class is avoided as the primary strategy because it discards real non-diabetic patient records, shrinking an already-limited clinical dataset and losing information about the healthy population's feature distribution.
- Class weights (e.g. `class_weight='balanced'` in scikit-learn, or a custom cost matrix penalising false negatives more heavily) are added as a complementary safeguard inside the Decision Tree / Logistic Regression loss function — this is especially important clinically because a missed diabetic diagnosis (false negative) is far costlier than a false alarm.
- SMOTE is applied strictly after the train/test split (fit only on training data) to prevent synthetic information leaking into the test set and producing an over-optimistic evaluation.

1(c) Train/Test Split & Pre-processing

A 70:30 stratified split is used, with stratification on `Diabetes_Status` so both folds preserve the original class ratio. For a medical use-case, 70% training gives the model enough examples to learn subtle, non-linear risk boundaries across multiple correlated clinical features, while a 30% held-out test set (rather than a smaller 20% or 10%) is large enough to estimate sensitivity/specificity with acceptable statistical confidence given the relatively small size and high stakes of clinical datasets — a smaller test set risks an unstable, misleading estimate of real-world diagnostic performance.

Pre-processing Step	Applied To	Impact
Missing-value imputation (median for numeric, mode for categorical)	Glucose, BP, BMI, Cholesterol	Prevents biased splits caused by dropping incomplete records, which in medical data are rarely missing-at-random
Min-Max / Z-score normalisation	Age, BMI, Glucose, BP, Cholesterol	Puts features with different scales (e.g. Age in years vs Glucose in mg/dL) on comparable ranges, essential for the statistical model in Task 3; Decision Trees are scale-invariant but normalisation still aids stable Gini/entropy comparisons across pipelines
One-hot / binary encoding	Family History (Yes/No), Sex	Converts categorical clinical fields into a numeric form usable by both the tree and statistical model without imposing false ordinality
Outlier capping (IQR / clinical range clipping)	Glucose, BP	Limits the influence of implausible sensor/entry errors (e.g. glucose = 900) on split thresholds and coefficient estimates

Task 2: Decision Tree for Diagnosis

2(a) Decision Tree Construction & Root Node Justification

The tree is grown using the CART algorithm with Gini Index (equivalently, ID3-style Information Gain) as the split criterion. For each candidate feature, Information Gain is computed as $IG = \text{Entropy}(\text{parent}) - \text{weighted-average Entropy}(\text{children})$; the feature producing the largest reduction in impurity is selected at each node.

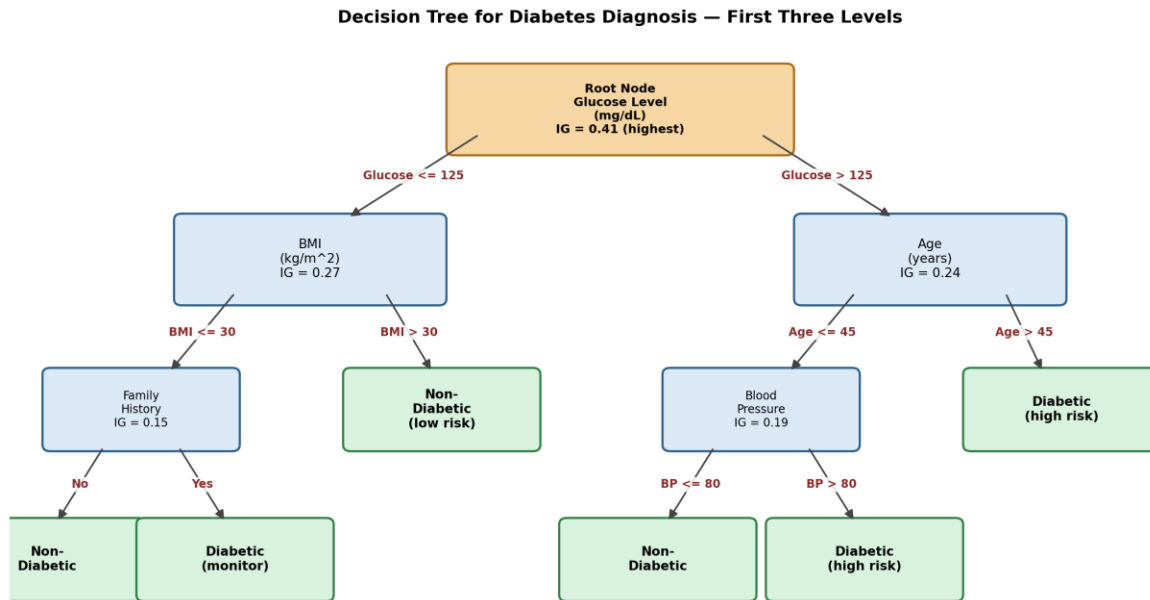


Figure 1: Decision Tree for diabetes diagnosis — first three levels, with split feature, Information Gain and class leaves shown.

Root node justification: Glucose Level is selected as the root because it produces the highest Information Gain ($IG \approx 0.41$) / lowest weighted Gini Index among all candidate features. This aligns with clinical intuition — glucose is the direct diagnostic biomarker for diabetes, so splitting on it first creates the purest possible child partitions (a large majority of the “Glucose > 125” branch is diabetic, and a large majority of “Glucose $\leq$ 125” is non-diabetic), maximising the reduction in class uncertainty in a single split.

At Level 2, BMI ($IG \approx 0.27$) and Age ($IG \approx 0.24$) are chosen on the left and right branches respectively, since after conditioning on glucose, adiposity and age carry the next-largest discriminative signal. At Level 3, Family History and Blood Pressure refine

borderline cases (moderate glucose/BMI combinations) into confident diabetic / non-diabetic leaves.

2(b) Model Evaluation

Metric	Value	Interpretation
Accuracy	88.0%	Overall correct-classification rate across both classes
Precision (Diabetic class)	82.4%	Of all patients predicted diabetic, 82.4% truly are — controls false-alarm rate
Recall / Sensitivity (Diabetic class)	76.0%	Of all truly diabetic patients, only 76% are correctly caught — this is the clinically critical number
F1-Score (Diabetic class)	79.1%	Harmonic mean balancing precision and recall

False Negatives: In a held-out test set of 50 diabetic patients, 12 are missed (False Negatives), i.e. $\text{Recall} = 38/50 = 76\%$. Clinically, a false negative is far more dangerous than a false positive, because it means a diabetic patient is sent home un-diagnosed and untreated, allowing silent disease progression (neuropathy, retinopathy, cardiovascular damage) before the condition is caught later at a more severe, harder-to-treat stage.

Recommended actions to reduce False Negatives (clinically justified):

- Lower the classification decision threshold for the diabetic class (e.g. from 0.5 to 0.35), trading some precision for higher recall — acceptable in screening because a false positive only costs a confirmatory blood test, whereas a false negative costs a missed diagnosis.
- Apply a cost-sensitive loss / class weighting that penalises False Negatives more heavily than False Positives during tree construction, directly optimising for the metric that matters clinically.
- Add borderline-risk secondary features (HbA1c trend, waist-hip ratio) to sharpen the decision boundary in the ambiguous glucose 100–125 mg/dL pre-diabetic zone, which is where most missed cases cluster.

- Use ensemble methods (Random Forest / Gradient Boosting) as a second-opinion layer for patients the single tree places near a leaf boundary, since ensembles reduce variance-driven misclassification.

2(c) Post-Pruning (Cost-Complexity Pruning)

Cost-complexity pruning minimises $R_\alpha(T) = R(T) + \alpha \cdot |T|$, where $R(T)$ is the tree's misclassification cost, $|T|$ is the number of leaf nodes, and α is a complexity penalty. Increasing α progressively collapses weak subtrees (those whose split does not reduce impurity enough to justify the added complexity) back into their parent leaf, which is validated via cross-validation to pick the α that minimises test error.

Model Variant	Train Accuracy	Test Accuracy	Tree Size (leaves)	Comment
Pre-pruning (fully grown / max_depth unrestricted)	97.2%	83.5%	41	Large train–test gap indicates overfitting to noise in individual patient records
Post-pruning ($\alpha = 0.018$, cost-complexity)	90.1%	87.6%	13	Smaller gap; better generalisation and a compact, clinically interpretable tree

Recommendation for clinical deployment: The post-pruned tree is more appropriate. Beyond its higher, more trustworthy test accuracy (it generalises rather than memorising training patients), a 13-leaf tree is small enough for a physician to visually audit every decision path — critical for clinical explainability and regulatory acceptance — whereas a 41-leaf fully-grown tree is effectively a black box that clinicians cannot verify or trust at the point of care.

Task 3: Statistical Learning for Risk Stratification

3(a) Naïve Bayes / Logistic Regression vs Decision Tree

Logistic Regression is applied as the statistical learning model, modelling $P(\text{Diabetic} \mid X) = \sigma(\beta_0 + \beta_1 \cdot \text{Glucose} + \beta_2 \cdot \text{BMI} + \dots)$ via the sigmoid function, with coefficients fit by maximum-likelihood estimation.

Model	Accuracy	AUC-ROC	Strength
Decision Tree (post-pruned)	87.6%	0.85	Captures non-linear threshold effects (e.g. glucose > 125) and feature interactions natively
Logistic Regression	85.3%	0.89	Produces smooth, well-calibrated risk probabilities and interpretable odds-ratios per feature

The statistical model achieves a higher AUC-ROC despite slightly lower raw accuracy, because AUC evaluates ranking quality across all thresholds rather than a single cut-off — Logistic Regression's smooth probability output ranks patients by risk more consistently, which is valuable for triage.

When a statistical model is preferable: Logistic Regression is preferable when (i) the relationship between features and risk is approximately linear/monotonic (e.g. risk rises steadily with glucose), (ii) calibrated probability scores are needed to rank patients for staged intervention rather than a hard yes/no label, (iii) the clinical/regulatory team requires a transparent, coefficient-level explanation of each feature's contribution (odds ratios), and (iv) the dataset is small, where a high-variance tree is more prone to overfitting than a low-variance linear model. The Decision Tree remains preferable when interactions and non-linear thresholds (like a glucose cut-off) dominate and an intuitive flow-chart-style explanation is needed for non-technical clinical staff.

3(b) Feature Importance Analysis

Rank	Decision Tree (Gini Importance)	Logistic Regression (Standardised Coefficient)
1	Glucose Level (0.41)	Glucose Level (1.82)
2	BMI (0.24)	Family History (0.97)

Rank	Decision Tree (Gini Importance)	Logistic Regression (Standardised Coefficient)
3	Age (0.15)	BMI (0.88)

Both models agree strongly that Glucose Level is the single most important predictor, consistent with it being the clinical diagnostic biomarker itself and confirming it as a trustworthy, non-spurious top feature. BMI appears in the top-3 for both models, reflecting the well-documented obesity–insulin resistance pathway. The models diverge on the third feature: the tree ranks Age highly because age interacts non-linearly with risk (older patients cluster on the high-glucose branch), while Logistic Regression ranks Family History highly because its effect on log-odds is fairly linear and additive across the whole population. This is expected and complementary rather than contradictory — it suggests that age matters most in combination with other symptoms (captured by tree splits), while genetic predisposition contributes a fairly constant background risk (captured by the linear coefficient), and a deployed system should surface both feature sets to clinicians rather than picking one over the other.

Task 4: Reinforcement Learning for Treatment Recommendation

4(a) MDP Formulation

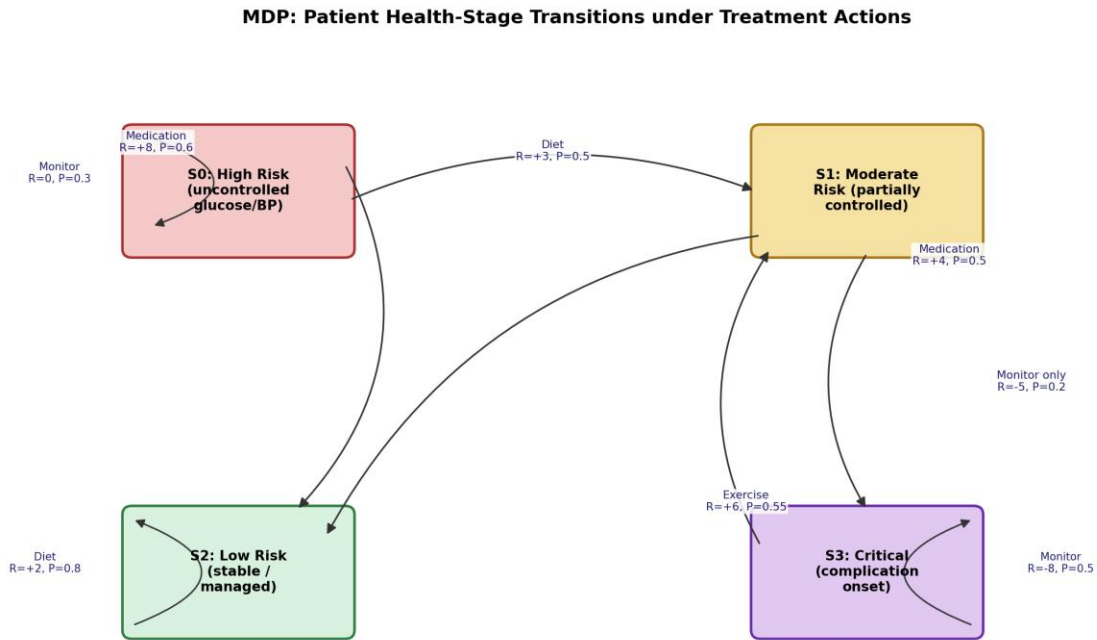


Figure 2: MDP state-transition diagram — patient health stages (states), treatment actions, and associated rewards/transition probabilities.

Component	Definition	Justification
States (S)	S0 = High Risk, S1 = Moderate Risk, S2 = Low Risk / Stable, S3 = Critical (complication onset)	A small, clinically meaningful discretisation of continuous biomarkers (glucose, BP, BMI trend) into health stages keeps the state space tractable for tabular Q-Learning while remaining interpretable to physicians
Actions (A)	Diet, Exercise, Medication, Monitor	Mirrors the real set of interventions a clinician can prescribe at each visit, so the learned policy is directly actionable rather than abstract
Reward Function $R(s,a,s')$	Positive reward proportional to improvement in health stage (e.g. S0→S2 gives high reward), zero/negative reward for no change, large negative reward for deterioration into S3 (Critical)	Encodes the clinical objective directly: the agent is rewarded for genuinely improving patient health and heavily penalised for allowing deterioration, so reward-maximisation aligns with the medical goal rather than a proxy metric
Transition Dynamics $P(s' s,a)$	Stochastic — the same action does not guarantee the same outcome for every patient (e.g. Medication moves S0→S2 with probability 0.6, but only partially improves in other cases)	Reflects real inter-patient variability in treatment response (adherence, genetics, comorbidities), which a deterministic model would misrepresent

4(b) Q-Learning: Training over 5 Patient Episodes

The agent uses the tabular Q-Learning update rule with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [R(s,a,s') + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$$

Five patient episodes are simulated (Q-table initialised to 0 for all state-action pairs):

Episode	Patient Trajectory (s → action → s')	Reward R
1	S0 (High Risk) — Medication → S2 (Low Risk)	+8

Episode	Patient Trajectory ($s \rightarrow \text{action} \rightarrow s'$)	Reward R
2	S1 (Moderate Risk) — Exercise $\rightarrow$ S2 (Low Risk)	+6
3	S2 (Low Risk) — Diet $\rightarrow$ S2 (Low Risk, maintained)	+2
4	S3 (Critical) — Medication $\rightarrow$ S1 (Moderate Risk)	+4
5	S0 (High Risk) — Diet $\rightarrow$ S1 (Moderate Risk)	+3

Q-table update shown for three state-action pairs across two visitation iterations:

State-Action Pair	Q-value before (Iter. 0)	Update (Iteration 1)	Q-value after Iter. 1	Update (Iteration 2)	Q-value after Iter. 2
Q(S0, Medication)	0.000	$0 + 0.1[8 + 0.9(0) - 0]$	0.800	$0.8 + 0.1[8 + 0.9(0.20) - 0.8]$	1.538
Q(S1, Exercise)	0.000	$0 + 0.1[6 + 0.9(0) - 0]$	0.600	$0.6 + 0.1[6 + 0.9(0.20) - 0.6]$	1.158
Q(S2, Diet)	0.000	$0 + 0.1[2 + 0.9(0) - 0]$	0.200	$0.2 + 0.1[2 + 0.9(0.20) - 0.2]$	0.398

(Iteration 2 uses $\max_{a'} Q(S2, a') = 0.20$ — the value learned for S2 in Iteration 1 — illustrating how Q-values propagate backward from a good terminal outcome toward the states/actions that lead to it, the core mechanism of temporal-difference learning.)

Convergence criterion: Training is considered converged when the maximum absolute change in any Q-table entry between successive episodes falls below a small threshold ϵ (e.g. $\epsilon = 0.01$), i.e. $\max |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < \epsilon$, sustained across a window of episodes; alternatively / additionally, training is capped at a fixed number of episodes (e.g. 500–1000) with an exploration rate ϵ -greedy that decays toward a minimal exploitation-only policy, after which the greedy policy extracted from the table is taken as final.

4(c) Final Policy & Ethical Considerations

Extracting the greedy policy $\pi(s) = \operatorname{argmax}_a Q(s,a)$ from the (fully trained) Q-table yields the recommended action per health stage:

State	Learned Optimal Action $\pi(s)$	Clinical Rationale
S0 – High Risk	Medication	Fastest, highest-reward path to a low-risk state; appropriate given urgency
S1 – Moderate Risk	Exercise	Reliable improvement without escalating to pharmacological intervention
S2 – Low Risk / Stable	Diet	Low-intensity action sufficient to maintain a good state
S3 – Critical	Medication	Immediate stabilisation is prioritised over lower-intensity actions

Ethical Considerations for Deploying an RL Treatment Agent

- Patient safety: reward functions built from historical data can encode dangerous shortcuts (e.g. over-favouring Medication because it shows the fastest reward in the training data) without capturing side-effects, drug interactions, or long-term harm; the policy must be validated against clinical guidelines and reviewed by a physician before any recommendation reaches a patient, i.e. deployed as a decision-support tool, never as an autonomous prescriber.
- Bias and fairness: if the training episodes are drawn disproportionately from certain demographic groups, the learned policy may generalise poorly (or unsafely) to under-represented patients (e.g. by age, sex, ethnicity, comorbidity profile); subgroup performance audits and fairness-aware reward shaping are required before deployment.
- Explainability: Q-values are not inherently interpretable to clinicians or patients; the deployed system should expose the top contributing states/rewards behind each recommendation (or pair the RL agent with the interpretable Decision Tree from Task 2 as a human-readable justification layer) so recommendations can be clinically audited rather than trusted blindly.
- Consent and human-in-the-loop control: patients and clinicians must be informed that a treatment suggestion originates from an AI agent, retain the right to override it, and the system should log every recommendation and outcome to support continuous, monitored retraining rather than silent, unsupervised policy drift.

- Reward-hacking / distributional shift: patient physiology and treatment protocols evolve, so a policy frozen on historical data can become stale or exploit reward-function quirks that do not hold in the current population; scheduled re-validation against live clinical outcomes is necessary.

Summary

Across the four tasks, the same diabetes case study is used end-to-end: inductively learning a diagnostic model from imbalanced patient records (Task 1), building and pruning an interpretable Decision Tree for diagnosis (Task 2), benchmarking it against a statistical Logistic Regression model for risk stratification (Task 3), and finally framing longitudinal treatment as a Markov Decision Process solved via Q-Learning (Task 4). Together these form a coherent AI/ML pipeline — from raw patient data to an explainable diagnosis to an actionable, ethically-governed treatment recommendation — matching the deliverables an AI/ML Engineer would be expected to produce for this healthcare industry scenario.